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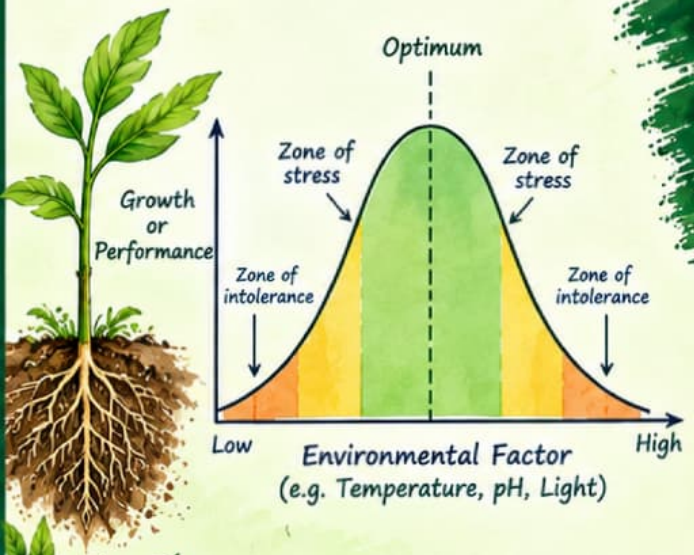
Law of Tolerance and Phytogeography

B.Sc. 2nd Semester | CBPBU

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Law of Tolerance

The ability of an organism to survive and function within a range of environmental conditions.



Phytogeography

The study of the geographical distribution of plants and vegetation on Earth.



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∅ Theory of Tolerance

The theory of tolerance was proposed by victor shelford in 1913. It states that the distribution, growth, reproduction and survival of an organism depend on its tolerance to various environmental factors such as temperature, light, water salinity, pH & humidity.

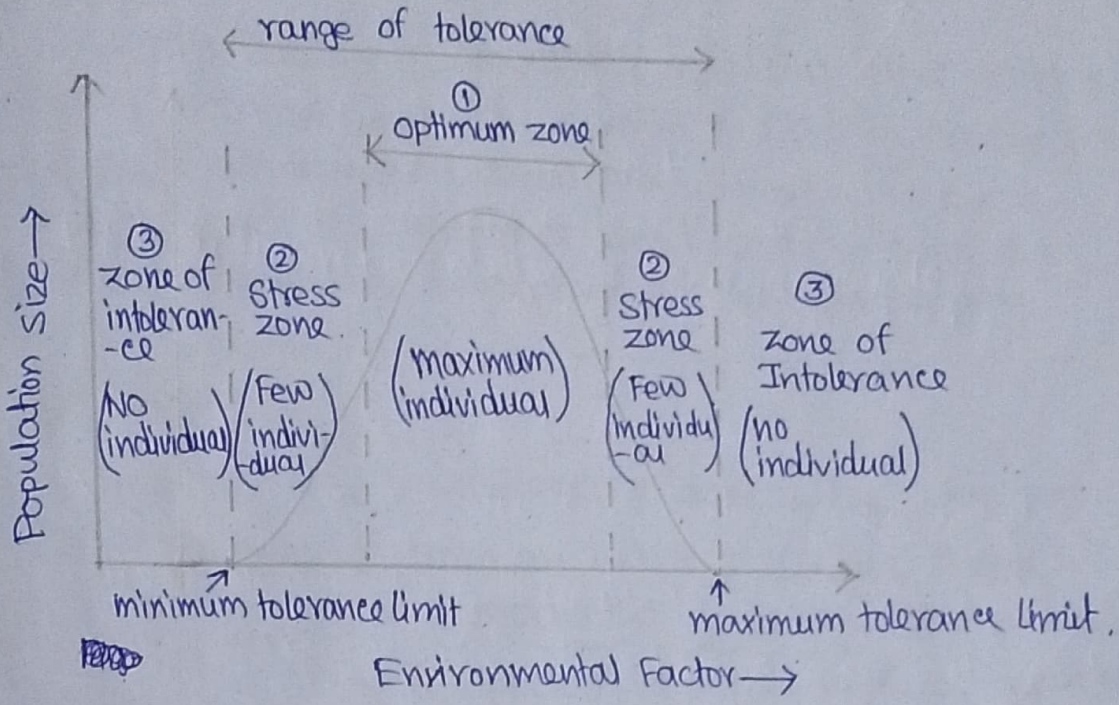


Fig- Graphical representation of shelford's Theory of Tolerance.

~~Princip~~ principles -

① Every organisms has a minimum tolerance limit & a maximum tolerance limit for each environmental factor.

② Between these two limit the optimum zone lies ~~which~~ where organism grows & reproduces most successfully.

③ The areas near the maximum & ~~minm~~ minimum limits are called as zone of stress where growth & reproduce are reduced.

④ If an environmental factor goes below the minimum tolerance limit or above the maximum tolerance limit the organism cannot survive.

⑤ The environmental factor nearest to the tolerance limit act as an limiting factor which restricting the distribution of the organism.

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Significance

① Explains the geographical distribution of plants and animals
② Helps to understand how different organisms adopt to different environment.
③ useful in ecology, agriculture & wildlife conservations.

Conclusion

Shelford's Theory of Tolerance explains that every organism can survive only within a certain range of ~~tolerance~~ environmental conditions. If any factor go ~~beyou~~ beyond the tolerance limit will restrict the growth, reproduction & survival of that organism.

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Phytogeography

Phyto + Geo + Graphy
↓ ↓ ↓
Plant earth Study/description

Phytogeography is the branch of Botany that studies the geographical distribution of plants and the factors affecting their distribution.

Factor affecting plant distribution-

CuSToMIsed → इतिहास = History
climate
Soil
Topography
मानव = Human
→ memory only.

① Climate — Temp^{n}, rainfall, Humidity
② Soil conditions - pH, salinity, minerals
③ Topography - Slope, Altitude
④ Human activities
⑤ Historical events.

Branches of Phytogeography

Descriptive
↓
where plant occur

Interpretive
↓
why plant occur in those areas.

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Principles of phytogeography –

① Plant Distribution–

Plants are not uniformly distributed uniformly throughout the world. Different environmental conditions, which support different types of vegetation.
ex– Cactus are common in deserts, while mangroves are found in coastal saline areas.

> Plant distribution varies from region to region depending on environmental suitability.

② Climatic influence–

Climate is the most important factor controlling plant ~~destr~~ distribution. Temperature, rainfall, humidity, sunlight & wind directly affect plant growth, reproduction & survival.

ex–
High rainfall → Tropical rainforest
Moderate rainfall → Grasslands.
Low rainfall → Deserts.
Low temperature → Coniferous forest.

③ Edaphic influence–

The physical & chemical ~~influence~~ properties of soil influence plant distribution. Soil texture, pH, mineral, organic matter moisture content, aeration determine ~~plant~~ which plant grow succuessfully.

ex–
Saline soil supports → Halophytes
Sandy soil supports → Xerophytes
Fertile soil supports → agricultural crops.

④ principle of Migration–

Movement + Establishment = Migration

Movement and successful establishment of a plant species after dispersal is called migration.

Types–
wind dispersal
water dispersal
Animal dispersal
Human dispersal
} + Successful establishment ⇒ migration

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⑤ principle of dispersal-

Movement of seeds, spores or propagules from one place to another is clo dispersal.

Types-

wind dispersal
water dispersal
Animal disperal
Human mediatld dispersal.

⑥ Principle of endemism-

Some plant species are restricted to a particular geographical area and are not naturally found elsewhere, such species are clo as endemic species

ex- Cycas beddomei → restricted to Eastern Ghats of Andhra Pradesh.

Strobilanthes kunthiana → restricted to Western Ghats
↳ Famous for blooming once in 12 year.

⑦ Tectonic plate movement & Continental drift-

The present distribution of plant families & are not the results of climate alone. The movement of land mass (continental drift) played massib massive role.

⑧ Ecological adaptation-

plants can survive only if they are adapted to the environmental conditions of their habitat. Adaptation may be anatomical, physiological & morphological.

ex-

Hydrophyte → possess aerenchyma
Halophyte → tolerate high saline condition
Xerophyte → possess thick cuticles & sunken Stomata.

⑨ # Theory of Tolerance- ⊛

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Dispersal v/s migration-

Feature	Dispersal	migration
① Definition	movements of seeds, spores from one place to another.	movement & successful establishment of a plant species to a new area.
② process	only transfer or transport	Tranfer followed by colonization & survival.
③ requirement	~~Des~~ Doesnot ~~a~~ require establishment in a new habitat	require successful establishment in a new habitat.
④ Stage	First stage in ~~plat~~ plant spread.	Later stage after dispersal.
⑤ Success need	No	Yes.
⑥ example	Cotton seed carried by wind to a new location	Cotton seed carried by wind to a new location & ~~germinati~~ germinated seed adapt to that locatio

Endemism

Endemism is the phenomenon in which a plant or animal species is naturally restricted to a particular geographical area and is not found naturally elsewhere in the world.

The species confined to a specific region ~~is~~ are called endemic species.

Types of Endemism

- Paleoendemism
- Neoendemism
- cytogenetic endemism
- Ecological endemism

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① Paleoendemism -

Paleoendemics are ancient species that were once widely distributed but are now confined to a limited geographical area due to climate change, competition or extinction in other regions.

example

Cycus revoluta

② Neoendemism -

Neoendemics are newly evolved species that are restricted to a particular area & are not found elsewhere.

example

Strobilanthus kunthiana

~~③ Cytoendemism -~~

③ ~~Cytogenic~~ Cytogenetic endemism -

Cytogenic endemism arises when chromosomal changes lead to the formation of a new endemic species.

example -

many grass species originated by chromosome doubling.

④ Ecological endemism -

Ecological endemics are the species restricted to a particular habitat as they can survive only under those specific environmental conditions.

ex -

Halophytes adapted to saline soil environment.

Xerophytes adapted to sandy soil environment.

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